

COMPOSITE MATERIAL DATASHEETS CORE ROHACELL HERO

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	ROHACELL HERO
Commercial name	EVONIK ROHACELL 51 HERO / 71 HERO / 110 HERO
Specifications	AIMS04-11-011 (Codes A, B & C)
Description	Rigid Foams, close cell, high elastic Polymethacrylimide (PMI)
Version	1.2

REFERENCES

- TEA-F9-MM-200001 Is. B "Design Values for rigid, closed-cell, high-elastic polymethacrylimide Rohacell® 71 Hero foam material"
- TE-T-NT-230036 Is. A "Derivation of Material Design Values for Rigid Foam Rohacell® 51 Hero".
- TE-T-NT-230064 Is. A "Derivation of Material Design Values for Rigid Foam Rohacell® 110 Hero".

AIMS04-11-011 code A (ROHACELL 51 HERO)

PHYSICAL PROPERTIES

Cell size	- mm
Density	52 kg/m ³
Max. Operational Limit (MOL)	>90 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..120°C
	α	-	-	-

ELASTIC MODULI

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	90°C/Dry	90°C/W		70°C/W	90°C/W	-55°C/Dry	RT/Dry	70°C/W	90°C/W
<i>E</i> (**)	35.8	30.7	20.0 (*)	26.2	19.4	0.80	0.651	0.632	28.6	24.6	16.0	15.5
<i>G</i>	28.4	22.8	13.9 (*)	17.3	13.0	0.80	0.609	0.570	22.7	18.2	11.1	10.4
ν	0.25					-	-	-	-	-	-	-

(*) Value estimated by linear interpolation.

(**) E_c tests results are the only ones considered for *E* design values definition.

STRENGTH

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	90°C/Dry	90°C/W		70°C/W	90°C/W	-55°C/Dry	RT/Dry	70°C/W	90°C/W
F_{3t}	2.0	2.2	1.5 (*)	1.4	1.5	0.71	0.722	0.681	1.42	1.56	1.06	1.06
F_{3c}	0.8	0.6	0.3 (*)	0.5	0.3	0.82	0.599	0.500	0.65	0.49	0.24	0.24
F_{su}	0.9	0.8	0.5 (*)	0.6	0.5	0.80	0.662	0.625	0.72	0.64	0.40	0.40

(*) Value estimated by linear interpolation.

OTHER PROPERTIES

Wrinkling parameter to consider in wrinkling strength analysis with this foam core, per formulation per CMH-17, Vol. 6, §4.6.6.2, shall be: $Q = 0.6$

COMPOSITE MATERIAL DATASHEETS CORE ROHACELL HERO

AIMS04-11-011 code B (ROHACELL 71 HERO)

PHYSICAL PROPERTIES

Cell size	- mm
Density	75 kg/m ³
Max. Operational Limit (MOL)	>90 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..120°C
	α	4.3E-5	2.5E-5	1.4E-5

ELASTIC MODULI

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)		
	-55°C/Dry	RT/Dry	70°C/W	90°C/W		70°C/W	90°C/W	RT/Dry	70°C/W	90°C/W
<i>E</i>	54.7	44.5	31.4	28.0	0.85	0.706	0.629	37.8	26.7	23.8
<i>G</i>	49.4	32.9	24.5	22.5	0.85	0.745	0.684	28.0	20.8	19.1
<i>v</i>	0.25				-	-	-	-	-	-

STRENGTH

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)		
	-55°C/Dry	RT/Dry	70°C/W	90°C/W		70°C/W	90°C/W	RT/Dry	70°C/W	90°C/W
<i>F_{3t}</i>	-	1.21	0.87	0.85	0.72	0.719	0.702	0.87	0.63	0.61
<i>F_{3c}</i>	-	0.87	0.56	0.52	0.83	0.644	0.598	0.72	0.46	0.43
<i>F_{su}</i>	-	0.94	0.67	0.65	0.83	0.713	0.691	0.78	0.56	0.54

OTHER PROPERTIES

Wrinkling parameter to consider in wrinkling strength analysis with this foam core, per formulation per CMH-17, Vol. 6, §4.6.6.2, shall be: $Q = 0.6$

COMPOSITE MATERIAL DATASHEETS CORE ROHACELL HERO

AIMS04-11-011 code C (ROHACELL 110 HERO)

PHYSICAL PROPERTIES

Cell size	- mm
Density	110 kg/m ³
Max. Operational Limit (MOL)	120 °C

CTE (1/°C)	Range	-55°C to 180°C
	α	$33.7 \cdot 10^{-6}$

ELASTIC MODULI

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>			B-basis Values (MPa)				
	-55°C/Dry	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	90°C/W	120°C/W
<i>E</i>	142.5	129.8	75.7	71.4	63.7	0.76	0.726	0.746	0.746	108.3	98.6	57.5	54.2	48.4
<i>G</i>	71.0	64.8	35.8	32.4	26.4	0.76	0.699	0.707	0.707	53.9	49.2	27.2	24.6	20.0
<i>v</i>	0.25					-	-	-	-	-	-	-	-	-

STRENGTH

Property	Mean Values (MPa)					B-basis K_B	<i>E-KDF</i>			B-basis Values (MPa)				
	-55°C/Dry	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	90°C/W	120°C/W
<i>F_{3t}</i>	4.7	5.1	2.8	2.6	2.0	0.73	0.774	0.839	0.839	3.4	3.7	2.0	1.8	1.4
<i>F_{3c}</i>	3.3	2.9	1.3	1.1	1.0	0.72	0.571	0.550	0.550	2.3	2.0	0.9	0.8	0.7
<i>F_{su}</i>	1.9	2.3	1.2	1.1	0.8	0.73	0.700	0.733	0.733	1.3	1.6	0.8	0.8	0.5

OTHER PROPERTIES

Wrinkling parameter to consider in wrinkling strength analysis with this foam core, per formulation per CMH-17, Vol. 6, §4.6.6.2, shall be: $Q = 0.6$